INVERSE FUNCTIONS QUIZ

BLAKE FARMAN

Lafayette College

Name: Solutions

Solve the following equations for x.

1. $\log_4(x+3) + \log_4(x-3) = 2$

$$2 = \log_4(x+3) + \log_4(x-3) = \log_4((x+3)(x-3)) = \log_4(x^2-9)$$

$$\Rightarrow \quad 4^2 = 4^{\log_4(x^2-9)} = x^2 - 9$$

$$\Rightarrow \quad x^2 = 16 + 9 = 25$$

$$\Rightarrow \quad \boxed{x = 5} \quad (\log_4(x+3) \text{ and } \log_4(x-5) \text{ are not defined at } -5).$$

2. $e^x = \left(\dfrac{e^{\sqrt{x}}}{e^{\frac{1}{2}}}\right)^2$

$$e^x = \frac{e^{2\sqrt{x}}}{e^{2(\frac{1}{2})}} = e^{2\sqrt{x}-1}$$

$$\Rightarrow \ln(e^x) = \ln(2\sqrt{x}-1)$$

$$\Rightarrow \quad x = 2\sqrt{x} - 1$$

$$\Rightarrow \quad x + 1 = 2\sqrt{x}$$

$$\Rightarrow \quad (x+1)^2 = (2\sqrt{x})^2 = 4x$$

$$\Rightarrow \quad 0 = x^2 + 2x + 1 - 4x = x^2 - 2x + 1 = (x-1)^2 = 0 \quad \Rightarrow \boxed{x = 1}$$